

Mariam Mohammed Elsoufy

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Experience

SAIB

Aug 2025 – Sep 2025

Core Banking Development Intern

Cairo, Egypt

- Developed a device inventory management system for tracking bank-issued assets, Designed and implemented database schemas, Enums, and relationships for devices, users, and actions
- Explored Temenos T24 core banking system

Enppi

Aug 2024 – Sep 2024

IT & Security Intern

Cairo, Egypt

- Configured Cisco switches, developed low-code web applications, and worked on Python-based data analysis and visualization.

Projects

IMMERSA – Interactive AR Historical Experience (Graduation Project) | *AI Voice Backend, LLMs, FastAPI, Real-Time Streaming*

- Designed and built the AI voice interaction backend for an AR time-travel experience, enabling real-time spoken Q&A with historical characters rendered on standalone AR headsets.
- Implemented a 5-stage async pipeline (audio preprocessing - STT - FAQ/LLM - TTS - WebSocket streaming) using FastAPI, achieving <1.5s latency from end-of-utterance to first audio output.
- Built an in-memory semantic FAQ cache using sentence-transformers embeddings and NumPy cosine similarity, reducing FAQ lookup from 50–500ms (PostgreSQL pgvector) to <1ms.
- Engineered TTS sentence pipelining with ElevenLabs, firing the next sentence's synthesis before the current one finishes streaming, eliminating inter-sentence gaps for seamless character speech.
- Integrated a parallel AI verifier (OpenAI) running concurrently with TTS to validate historical accuracy, character consistency, and modern-reference detection — capable of aborting the stream mid-playback on failure.
- Implemented multi-layer content filtering (English + Arabic) and per-character fallback audio, ensuring the user never experiences silence on any failure path.

Spotify Music Recommender | *Machine Learning, PySpark, Recommendation Systems*

- Built a hybrid music recommendation system using PySpark, combining content-based filtering and collaborative filtering across 738K+ songs and 11.3M user-playlist interactions.
- Implemented content-based recommendations using LSH-accelerated cosine similarity over 8 audio features (danceability, energy, valence, etc.), blending similarity-ranked songs with same-artist picks.
- Processed and merged multiple Spotify and Million songs Dataset sources into a unified pipeline spanning EDA, preprocessing, modeling, and evaluation notebooks.

TawasolApp – LinkedIn Clone | *Flutter, Dart, REST APIs, Stripe*

- Built Developed a LinkedIn Android clone using Flutter and Dart with connections, search, privacy, and premium subscription features.
- Implemented connection workflows, follow/unfollow, block/unblock, and reporting systems.
- Built search for users, posts, jobs, and companies.
- Integrated Stripe payments with premium plan selection.

Virtual Controls Application | *Computer Vision, OpenCV, Python, Gesture Recognition*

- Built a gesture recognition Windows Application from scratch using Python and OpenCV.
- Implemented fingertip detection via convexity defects and hand orientation using geometric angles.
- Mapped gestures to commands using finger count and hand direction to Computer controls such as: copy, paste, cut, volume up, volume down, etc.

Malware Detector | *Python, YARA, Cybersecurity*

- Developed a Python-based malware detection system using custom YARA rules to identify ransomware signatures in executable files.
- Implemented a rule-based scoring system with configurable thresholds to classify files as malicious or safe.

Education

Cairo University – STEP

Sep 2021 – Jun 2026

Bachelor of Science in Computer and Communication Engineering | Grade: Very Good